Complete at least 2 exercises.

Throughout, $\Bbbk$ is a field, and all modules are right modules unless a left action is specified.

Exercise 1.

(1) Let $A = \bigoplus_{d\geq 0} A_d$ be a graded algebra with $S = A_0$ finite-dimensional semisimple.

i. Prove that $A/A_+ \cong S$ and $\operatorname{rad} A \subset A_+$.

ii. Prove that if $A_+^N = 0$ for some $N \geq 1$ (for example if $\dim_\Bbbk A < \infty$), then $\operatorname{rad} A = A_+$.
Hint. Argue as for the strictly upper triangular ideal $J \subset \Bbbk A_n$ in the lecture.

iii. Show that for $A = \Bbbk[x]$, $\deg x = 1$, one has $\operatorname{rad} A = 0 \neq A_+$.

iv. Suppose that $\dim_\Bbbk A < \infty$ and that A is generated in degree 1 over S. Prove that

$$\operatorname{rad}^k A = \bigoplus_{d\geq k} A_d, \qquad \operatorname{rad}^k A/\operatorname{rad}^{k+1} A \cong A_k \qquad (k \geq 0).$$

(2) Let $A = \Bbbk A_3$, $M = e_{22}A$, and consider $\mathbb{D}(M)_d = \operatorname{Hom}_\Bbbk(M_{-d}, \Bbbk)$ with $(af)(m) = f(ma)$.

i. Find a homogeneous basis of $\mathbb{D}(M)$, its degrees, and all non-zero actions of matrix units on it.

ii. Show that $\Bbbk e_{13} \subset Ae_{33}$ is a graded left submodule and that $\mathbb{D}(M) \cong (Ae_{33}/\Bbbk e_{13})(1)$ as graded left A-modules.

iii. Deduce that $\mathbb{D}(M)$ is not isomorphic to any $(Ae_{jj})(r)$, $j \in \{1, 2, 3\}$, $r \in \mathbb{Z}$.

(3) Let $A = \Bbbk A_n$, $P_i := e_{ii}A$ with the inherited grading, and $L_j := e_{jj}A/e_{jj}A_+$ as graded module concentrated in degree 0.

i. Prove that for $k \geq 0$, $\operatorname{rad}^k P_i = \operatorname{span}_\Bbbk\{e_{im} \mid m \geq i + k\}$.

ii. Prove that for $0 \leq k \leq n - i$, $\operatorname{rad}^k P_i/\operatorname{rad}^{k+1} P_i \cong L_{i+k}(-k)$ as graded right A-modules, and deduce that

$$P_i \cong P_j(d) \text{ as graded modules} \iff (i, d) = (j, 0).$$

Exercise 2.

(1) Let $S = \bigoplus_{i=1}^{4} \Bbbk e_i$ be a semisimple algebra with coordinate idempotent e_i for $1 \le i \le 4$ and let $V = \Bbbk a \oplus \Bbbk b \oplus \Bbbk c \oplus \Bbbk d$ be the S-bimodule with

$$a \in e_1 V e_2, \qquad b \in e_2 V e_4, \qquad c \in e_1 V e_3, \qquad d \in e_3 V e_4,$$

and $e_i V e_j = 0$ for all other (i, j). For $\lambda \in \Bbbk$ put

$$A_\lambda = T_S(V)/(ab - \lambda cd).$$

i. Find $R^\perp$ and a presentation of $A_\lambda^!$, for every $\lambda \in \Bbbk$.

ii. Dualise once more: compute $(A_\lambda^!)^!$ and recover the relation $ab - \lambda cd$.

iii. Prove that $A_\lambda \cong A_\mu$ as graded algebras if and only if λ, μ are either both zero or both non-zero.
Hint. A graded isomorphism over S sends each of a, b, c, d to a scalar multiple of itself.

(2) Let $A = \Bbbk[x, y]/(x^2, y^3)$ with $\deg x = \deg y = 1$, and let $V = A_1$.

i. Compute $h_A(t)$.

ii. Let $R = \ker(V \otimes_\Bbbk V \to A_2)$ and $Q = T_\Bbbk(V)/(R)$. Show that

$$Q \cong \Bbbk[x, y]/(x^2), \qquad h_Q(t) = \frac{1 + t}{1 - t}.$$

iii. Prove that A is not quadratic, and find the smallest degree in which A and Q differ.

(3) Let $A = \Bbbk A_4/(e_{13})$ with the quotient grading, and put $a = e_{12}$, $b = e_{23}$, $c = e_{34}$.

i. Find S, V and $R \subset V \otimes_S V$ with $A \cong T_S(V)/(R)$, a basis of A, and $h_A(t)$.

ii. Compute $R^\perp$, a presentation and a basis of $A^!$, and $h_{A^!}(t)$. Verify

$$h_A(t)\, h_{A^!}(-t)^{\mathsf{T}} = I_4,$$

and check that $(A^!)^! \cong A$.

Exercise 3.

(1) For $n \ge 3$ and $Z \subset \{1, \dots, n-2\}$ let

$$A_Z = \Bbbk A_n/(e_{k,k+2} \mid k \in Z)$$

with the quotient grading.

i. Show that

$$\{e_{ii}\}_i \ \cup \ \{e_{ij} \mid i < j,\ \nexists\, k \in Z :\ i \le k \le j - 2\}$$

is a basis of A_Z.

ii. Write $A_Z = T_S(V)/(R)$ with explicit S, V, and $R \subset V \otimes_S V$.

(2) Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ and $V = \Bbbk a \oplus \Bbbk b$ with

$$a \in e_1 V e_2, \qquad b \in e_2 V e_1,$$

and $e_i V e_j = 0$ otherwise.

i. Find a basis of $T_S(V)_d$ for all $d \ge 0$, and compute $h_{T_S(V)}(t)$.

ii. Find a basis of $A = T_S(V)/(ab)$ and show that $\dim_\Bbbk A < \infty$.

(3) Let A be a locally finite-dimensional graded algebra generated in degree 1 over the finite-dimensional semisimple algebra $S = A_0$.

i. Fix a bimodule ${}_SV_S$ and an S-bimodule isomorphism $\varphi : V \xrightarrow{\sim} A_1$. Prove that

$$h_A(t) = h_{T_S(V)}(t) \quad \Longrightarrow \quad T_S(V) \xrightarrow{\sim} A,$$

where the isomorphism of graded algebras on the right-hand side is induced by id_S and φ.

ii. Consider $\Bbbk[x, z]/(x^2)$ with $\deg x = 1$ and $\deg z = 2$. Show that

$$h_B(t) = h_{T_\Bbbk(B_1)}(t) \qquad \text{but} \qquad B \ncong T_\Bbbk(B_1)$$

as graded algebras.

(4) Let

$$B = T_\Bbbk(\Bbbk x \oplus \Bbbk y)/(x^2), \qquad \deg x = \deg y = 1, \qquad a_d := \dim_\Bbbk B_d.$$

i. Find a basis of B_d and compute $a_0, \ldots, a_6$.

ii. Prove that

$$a_d = a_{d-1} + a_{d-2} \quad (d \geq 2), \qquad h_B(t) = \frac{1+t}{1-t-t^2}.$$

iii. Compute $B^!$ (with $X = x^*$, $Y = y^* \in D(V)$), find a homogeneous basis of $B^!$ and $h_{B^!}(t)$, and verify $h_B(t)\, h_{B^!}(-t) = 1$.

Exercise 4.

(1) Let $S = \bigoplus_{i=1}^{r} \Bbbk e_i$, let V be a finite-dimensional S-bimodule in degree 1, and put

$$C_{ij} = \dim_{\Bbbk} e_i V e_j, \qquad A = T_S(V)/(V \otimes_S V) = S \oplus V.$$

i. Prove that

$$A^! = T_S(D(V)), \qquad h_A(t) = I + tC, \qquad h_{A^!}(t) = (I - tC^{\mathsf{T}})^{-1}.$$

ii. Deduce that $h_A(t)\, h_{A^!}(-t)^{\mathsf{T}} = I$.

(2) For $q \in \Bbbk$ let

$$C_q = \Bbbk\langle x, y\rangle/(x^2,\ y^2,\ xy + qyx), \qquad \deg x = \deg y = 1.$$

i. Find a homogeneous basis of C_q and compute all products of two basis elements, treating $q = 0$ and $q \neq 0$. Justify linear independence of your basis.

ii. For $u = \alpha x + \beta y$ compute u^2, and determine

$$\{(\alpha, \beta) \in \Bbbk^2 \mid u^2 = 0\} \qquad \text{for each } q,$$

and the q for which every $u \in (C_q)_1$ satisfies $u^2 = 0$.

iii. Prove that

$$C_q \cong C_{q'} \text{ as graded algebras} \iff q = q' \text{ or } qq' = 1.$$

Deduce that $C_1 \ncong C_{-1}$ if $\operatorname{char} \Bbbk \neq 2$ (even ungraded), and say what changes if $\operatorname{char} \Bbbk = 2$.
Hint. A graded isomorphism is determined by a linear bijection of the degree-1 parts, and preserves $\{\Bbbk u \mid u^2 = 0\}$.

iv. Compute a quadratic presentation of $C_q^!$ (with $X = x^*$, $Y = y^* \in D(V)$), and $h_{C_q^!}(t)$. Verify that $h_{C_q}(t)\, h_{C_q^!}(-t) = 1$, and compare $C_0^!$ with $C_1^!$.